

Sorting Level 1 - Foundation Problems

1. Bubble Sorting

```
arr = [5, 3, 8, 1, 2]

def bubble_sort(arr):
    n = len(arr)

    for i in range(n):

        swapped = False
        for j in range(0, n - i - 1):

            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
                swapped = True

        if not swapped:
            break

    return arr

print(bubble_sort(arr))
```

2. Selection Sorting

```
arr = [5, 3, 8, 1, 2]

def selection_sort(arr):

    n = len(arr)

    for i in range(n):
        min_ind = i
        for j in range(i+1, n):
            if arr[j] < arr[min_ind]:
                min_ind = j

        arr[i], arr[min_ind] = (arr[min_ind], arr[i])

    return arr

print(selection_sort(arr))
```

3. Insertion Sorting

```
arr = [5, 3, 8, 1, 2]

def insertion_sort(arr):

    for i in range(1, len(arr)):

        key = arr[i]
        j = i - 1

        while j >= 0 and arr[j] > key:
            arr[j + 1] = arr[j]
            j -= 1

        arr[j + 1] = key

    return arr

print(insertion_sort(arr))
```